

phase3 system design

1. Objective of Phase 3

Upgrade system from:

“evaluate users individually”

to:

evaluate users in context of the network they form

2. What Phase 3 Adds

Core Additions

Graph construction (A_{ij}) Trust propagation (T_i) Coordination detection (group behavior) Graph-aware anomaly amplification

Still NOT included

ML models Full location system Memory similarity

3. Core Idea

A user is not judged only by:

their own behavior

but also by:

who they behave like

4. Graph Construction

4.1 Nodes

Each user = node i

4.2 Edge Definition

Create edge between users if:

they interact with same posts

4.3 Edge Weight

$$A_{\{ij\}} = w_1 \text{ Agree}_{\{ij\}} + w_2 \text{ TimeSim}_{\{ij\}} + w_3 \text{ Freq}_{\{ij\}}$$

Agreement

$$\text{Agree}_{\{ij\}} = (s_i = s_j)$$

Time Similarity

$$\text{TimeSim}_{\{ij\}} = e^{-\{ \}}$$

Frequency

$$\text{Freq}_{\{ij\}} =$$

5. Normalization

$$_{\{ij\}} =$$

6. Trust Propagation

6.1 Formula

$$T_i = \sum_j \{ij\} T_j + (1 - \sum_j \{ij\}) R_i$$

6.2 Iterative Solution

$$T = R \text{ repeat: } T_{\text{new}} = \lambda_g \cdot A_{\text{norm}} \cdot T + (1 - \lambda_g) \cdot R \text{ until convergence}$$

7. Updated User Weight

Phase 3 Weight

$$w_i = T_i \exp(-\text{Anom}_i)$$

Key Change

$$R_i \rightarrow \text{replaced by } T_i$$

8. Coordination Detection (Critical)

8.1 Pairwise Similarity

$$\text{Sim}(i,j) =$$

8.2 Coordination Score

$$S_{\{\text{coord}\}}(i) = \sum_j \text{Sim}(i,j)$$

8.3 Group Detection (Optional Upgrade)

$$\text{clusters} = \text{connected_components}(A_{ij} \text{ where weight} > \text{threshold})$$

8.4 Coordination Penalty

$$\text{Anom}_i^{\{\text{coord}\}} = S_{\{\text{coord}\}}(i)$$

Update anomaly:

$$\text{Anom}_i = 1 - (- (1 D_{\{\text{rate}\}} + 2 D_{\{\text{entropy}\}} + 3 S_{\{\text{coord}\}}))$$

9. Pipeline (Phase 3)

Vote received $\downarrow$ Update user: R_i , Exp_i , $Anom_i$ $\downarrow$ Update graph edges A_{ij} $\downarrow$ Run trust propagation $\rightarrow T_i$ $\downarrow$ Compute weight w_i $\downarrow$ Update post: N_j , S_{j+} , S_{j-} $\downarrow$ Compute: C_j , Var_j $\downarrow$ Alert decision

10. What Phase 3 Solves

Problem 1: Coordinated Bots

Many fake users voting same way

Before:

System fooled

Now:

High similarity $\rightarrow$ high anomaly $\rightarrow$ low weight

Problem 2: Trust Farming

Users behave well, then attack

Now:

Trust tied to neighbors $\rightarrow$ corrupted cluster collapses

Problem 3: Echo Chambers

Group reinforces itself

Now:

Graph normalization limits amplification

11. Backend Changes

New Table: Graph Edges

graph_edges (user_i TEXT, user_j TEXT, weight FLOAT, updated_at TIMESTAMP)

New Computation Job

every N seconds: run trust propagation

12. Optimization Strategy

Sparse Graph

keep only top-K neighbors per user

Incremental Update

update edges only for users involved in new interaction

Batch Propagation

run T_i computation every 30–60 seconds

13. Hyperparameters (Phase 3)

λ_g → trust propagation strength τ → time similarity scale α_3 → coordination anomaly weight K → max neighbors per node

14. What to Test

Test 1: Coordinated Attack

10 users vote same thing simultaneously

Expect:

$S_{\text{coord}} \uparrow \rightarrow \text{Anom} \uparrow \rightarrow w_i \downarrow \rightarrow \text{attack suppressed}$

Test 2: Honest Cluster

real users agreeing naturally

Expect:

moderate similarity → no heavy penalty

Test 3: Mixed Network

good + bad users connected

Expect:

bad users dragged down by anomaly good users maintain trust

15. Phase 3 Summary

Phase 2: “who you are matters” Phase 3: “who you behave like matters”

16. System Evolution

Phase Capability MVP basic credibility Phase 2 user-aware Phase 3 network-aware (robust)

17. Final Insight

This is the stage where your system becomes:

resistant to coordinated manipulation

Without Phase 3, any large-scale attack can still break the system.

Next Phase (Phase 4 Preview)

You will add:

Location + proximity Spatial trust Localized alerts

This introduces real-world context awareness.